Search and Competition in the Market for Sick Leaves

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Abstract

I develop and estimate a model of the market for medical treatment and sick leave, in which patients search imperfectly across physicians who differ in quality and choose how leniently to issue sick leave. Competition for patients leads lower-quality physicians to relax their issuance threshold, producing assortative matching between "dishonest" physicians and patients. I calibrate the model using data from Chile's national public insurer. In a counterfactual, a 64% increase in the fine for issuing sick leave reduces aggregate sick leaves by 32%, concentrated in the patients with the least medical need.

1 Introduction

Sickness insurance is subject to moral hazard. Sick workers should stay home, both for their own health and for firm productivity (Chunyu et al., 2026), but employers and insurers cannot verify illness, so healthy workers can also claim sick leave. The standard solution is third-party certification by physicians, who already diagnose and treat the patient and can therefore certify the illness at little extra cost.

Certification creates a second agency problem. When sick leave is paid by a public insurer, the physician acts as a gatekeeper whose income and professional duty both lie with the patient, while the cost falls on the insurer. Even absent financial motives, physicians have little reason to doubt unverifiable symptoms (Carlsen et al., 2020). When patients can choose their physician, the incentive to certify grows stronger, since lenient physicians attract more patients (Markussen and Røed, 2017).

Lenient issuance and sick leave fraud are a known issue in Chile, where workers in many cases have wide latitude in choosing their physician, either through a private insurer (ISAPRE) or through the free-choice modality (*Modalidad de Libre Elección*) of FONASA, the public insurer, which holds an 82% market share. In a 2024 survey of 1,200 adults in Chile's three most populous regions, 51% of respondents knew someone who obtained sick leave without being sick, and 56% believe buying one would be easy (Instituto de Salud Pública UNAB, 2024). In May 2025, after the Comptroller General found that about 25,000 public employees had travelled abroad while on sick leave, Congress passed Law 21.746, adding requirements for issuing physicians and raising penalties for over-issuance.[1][2] The administrative data for this paper only run through 2022, but it would be a natural extension in the future to calibrate the model with the use of this natural experiment.

*This paper is an edited and abridged version of my master's thesis (Tagle, 2024). The original version is available at leontagle.github.io, along with other projects that better represent my current interests. To save space, the list of references is omitted here; it is available in this link.

[1] See https://www.bcn.cl/leychile/Navegar?idNorma=1213603.

[2] Although the draft of this master's thesis was publicly available months before the sick leaves scandals broke out, I was not contacted by the government for proposals.

I present two models in which physicians play a dual role, as medical caregivers and as issuers of sick leave, and patients with heterogeneous preferences over these two roles search among them. Each physician chooses a level of leniency to maximize revenue net of a convex penalty in the number of sick leaves issued. Patients choose physicians based on the physician's quality of care, the price of a visit, and the probability of obtaining a sick leave, which depends on both the patient's medical need and the physician's leniency. In both models, physicians' leniency is public information.

To fix ideas, I first discuss a model where I shut down patient search: patients are randomly assigned a physician and they can only choose whether to visit her or not.[3] Each physician then acts as a local monopolist, choosing a threshold $\bar{\kappa}_j$ of medical need above which she grants sick leave. Her optimal threshold rises with her quality, because higher-quality physicians attract more patients who visit for care alone and so gain less from attracting patients with sick leave. Once this "captured clientele" is large enough, she jumps to the strictest admissible threshold, $\bar{\kappa}_{\max}$, which can be read as the most stringent medically sanctioned standard.

I then introduce patient search in two ways. In the first, patients search sequentially, as in McCall (1970): they draw physicians at random, one at a time and without recall, and visit the first whose expected utility exceeds their reservation utility. The second replaces this process with a logit discrete choice model, which approximates the first model while being simpler to compute. I call them, respectively, the "explicit" and "implicit" search models, or simply as the sequential and logit models.

With search, leniency becomes strategic: a physician who is too strict loses patients to more lenient competitors, so each physician's optimal threshold depends on everyone else's. I compute the equilibrium numerically with a best-response algorithm.

Finally, using administrative records of individual sick leaves from Chile's public insurer, which identify both the issuing physician and the patient, I estimate the model by SMM and run a policy counterfactual. Since physicians both diagnose patients and issue their leaves, a regulator cannot police individual cases of fraud, and so cannot implement the first best of denying leave to the least needy, and must instead rely on penalizing physicians' total leaves granted. I show that raising the fine on issuance by 64% achieves a 32% reduction in aggregate sick leave volume, concentrated on the patients with least medical need due to endogenous assortative matching between "dishonest" patients and physicians and general-equilibrium effects that induce all physicians to become stricter.

Literature Review. This paper studies patient choice and physician decision-making in a market setting. Studies in the literature have shown, on the one hand, how physicians respond to financial incentives (Clemens and Gottlieb, 2014, Robertson et al., 2012, Hennig-Schmidt et al., 2011, Brosig-Koch et al., 2016, 2017); on the other, how patients choose physicians based on what they observe (Bensnes and Huitfeldt, 2021), though subject to information frictions (Brown et al., 2023, Ginja et al., 2025), which I shut down in my models. Currie et al. (2023) bring the two together: physicians facing more competition shift their services toward what patients want. My models draw on Schnell (2026), who

[3] I'll be referring to physicians as "she", patients as "he", as needed.

studies prescription opioids with patient search. On top of allowing physicians to differ in quality as well as leniency, my main departure from her is the introduction of strategic interactions between physicians, who endogenously choose their certification threshold in response to other physicians' leniency.[4]

Oteíza (2024) studies sick leave issuance in Chile, using the 2021 suspension of electronic issuing rights for 188 high-issuing physicians as a quasi-experiment. Their issuance fell, but patients partly switched to other physicians, which dampened the aggregate effect. Markussen and Røed (2017) study the effect of competition on general practitioners' sick leave issuance in Norway, finding that GPs with a reputation for granting more sick absence can expect higher demand and GPs in turn adjust their issuance accordingly, raising aggregate absenteeism by roughly 3–4%. My models replicate this result qualitatively.

This paper is part of a wider literature on sick leave, which mostly studies how the rules of sickness insurance affect workers' absence and its effect on productivity (Johansson and Palme, 2005, Henrekson and Persson, 2004, Markussen et al., 2012, De Paola et al., 2014, Stearns and White, 2018), taking issuance as given. I study instead how issuance is determined upstream in a market between patients and physicians.

Finally, as noted above, this study joins a literature on the role of physicians as gatekeepers for social insurance (Markussen and Røed, 2017, Carlsen et al., 2020, Cabral and Dillender, 2025). Doctors' discretion shapes workers' access to benefits, and their incentives to prevent abuse are not aligned with the insurer's. In my models, the only constraint on over-issuance comes from a fine that is convex in the number of certificates a physician issues. In a counterfactual, I raise this fine to bring aggregate issuance down to a planner's target.

The rest of the paper proceeds as follows. Section 2 presents the model: a no-search benchmark and the two search specifications. Section 3 describes the algorithms used to compute equilibria for both models and illustrates with examples. Section 4 introduces the data on sick leave issuance. Section 5 shows the calibration results and discusses identification. Section 6 runs the penalty counterfactual, and Section 7 concludes.

2 Model

2.1 The Physician-Patient Market

I consider a market with $i = 1, ..., I$ patients and $j = 1, ..., J$ physicians. Patients are characterized by their "medical need" $\kappa_i \in \mathbb{R}^+$ and their "taste for sick leave" $\gamma_i \in \mathbb{R}$, drawn independently from the ex-ante cumulative distributions $F(\kappa)$ and $G(\gamma)$. Physicians are described by their "service quality" $V_j \in \mathbb{R}^+$, cost of visit $\tau_j \in \mathbb{R}^+$, and revenue function $R_j(\cdot)$. The distributions F and G and all physician characteristics are public knowledge.

A patient i can visit a physician for treatment and may be also granted sick leave. After the patient is matched with a physician j, his utility function, which is implicitly dependent

[4]This key difference between my model and Schnell's comes from physician utility that is not additively separable across patients, which is worked out in more detail in the Appendix of the thesis version of this paper (Tagle, 2024).

on his characteristic (κ_i, γ_i) tuple, is defined piecewise as follows:

$$U_i(V_j) = \begin{cases} \gamma_i + V_j\kappa_i - \tau_j & \text{if he is granted sick leave,} \\ V_j\kappa_i - \tau_j & \text{if he only visits the physician,} \\ 0 & \text{if he does not visit.} \end{cases}$$

Patient utility has three components: an interaction between the patient's medical need κ_i and the physician's service quality V_j, his taste for sick leave γ_i if he is granted one, and the cost of visit τ_j. The term $V_j\kappa_i$ is the value of treatment, where medical need and service quality are complements, and κ_i takes on the interpretation of the patient's marginal willingness to pay for physician quality.[5]

Whereas patients may visit at most one physician, a physician may see several patients. I define Q_j as the expected number of patients who will visit physician j. I say 'expected' because in my models patient visit will not be deterministic, they will match with physicians with a certain probability, and physicians choose their strategy ex-ante. Likewise I define X_j as the *expected* number of sick leaves granted by physician j, given her ex-ante patient demand and how many of them would be granted one given her chosen leniency.

I define the physician's utility function as follows:

$$U_j(Q_j, X_j) = R_j(Q_j) - P(X_j)$$

where the *revenue* function $R_j(\cdot)$ is physician-specific, increasing, and weakly concave in her expected number of patients, and the *punishment* function $P(\cdot)$ is increasing and strictly convex in her expected number of sick leaves. Physicians balance their utility from additional patients with what should be interpreted as the increased probability of a fine as they issue more sick leave. I assume both $R_j(\cdot)$ and $P(\cdot)$ are twice differentiable.

Following Schnell (2026), I focus on threshold equilibria, wherein each physician's strategy is the choice of a value $\bar{\kappa}_j \in \mathbb{R}_0^+$, such that of the patients who visit j, those with a κ_i value above or at that threshold will be granted sick leave, and those strictly under it will not. As a result of this, each physician will be identified by their given 'service quality', revenue function, cost of visit and choice of threshold: $\{V_j, R_j(\cdot), \tau_j, \bar{\kappa}_j\}$.[6] The set $\{\bar{\kappa}_j\}_{j=1}^J$ will be known to patients when searching for a physician.

The timing of this game is as follows:

- First stage: Physicians simultaneously choose $\bar{\kappa}_j$.
- Second stage: Observing the set $\{V_j, \tau_j, \bar{\kappa}_j\}_{j=1}^J$, each patient chooses (or is assigned to) some physician j.
- Third stage: Each patient can choose to see his physician and pay the fee τ_j; if he does not, he gets zero utility.

[5] Willingness to pay would also depend on the patient's wealth. To abstract from this, I interpret κ_i as the *average* willingness to pay for V_j.

[6] In this game, I take both $R_j(\cdot)$ and τ_j to be determined exogenously, fitting my setting and many others in which physicians' income and visit fees are highly regulated.

Conditional upon his visit, the utility of patient i from seeing physician j is:

$$U_i(V_j, \bar{\kappa}_j) = \begin{cases} \gamma_i + V_j\kappa_i - \tau_j & \text{if } \kappa_i \geq \bar{\kappa}_j, \\ V_j\kappa_i - \tau_j & \text{if } \kappa_i < \bar{\kappa}_j \end{cases}$$

I will usually use u_{ij} as a shorthand.

2.2 Non-Search Equilibrium

For illustration purposes, I first devote attention to a non-search baseline, where in the second stage, each patient is assigned to one of the J physicians uniformly at random, independently across patients; in the third, he only decides whether to visit her. Each physician thus faces the same pool of I/J patients in expectation and acts as a local monopolist.

A patient won't visit his assigned physician if his expected utility from the visit is negative; I call this the *free disposal* requirement. A physician j's expected patient demand as a function of her threshold $\bar{\kappa}_j$ and quality V_j is the following:

$$Q_j(\bar{\kappa}_j) \;=\; \frac{I}{J}\left[\int_{\tau_j/V_j}^{\infty} dF(\kappa) + \int_{\min\{\bar{\kappa}_j, \tau_j/V_j\}}^{\tau_j/V_j} \int_{\tilde{\gamma}(\kappa)}^{\infty} dG(\gamma)\, dF(\kappa)\right] \tag{1}$$

where $\tilde{\gamma}(\kappa) := \tau_j - V_j\kappa$. The first term consists of the mass of patients who just by virtue of physician j's service quality V_j would be willing to pay a visit (i.e. $\kappa_i \geq \tau_j/V_j$), and the second term would be patients who see physician j solely out of the expectation of getting sick leave ($\kappa_i \geq \bar{\kappa}_j$ and $\gamma_i \geq \tau_j - V_j\kappa_i$), and wouldn't visit otherwise.

Given that each patient with a κ_i higher or equal to $\bar{\kappa}_j$ is granted sick leave, the expected total number of such certificates granted by j, as a function of $\bar{\kappa}_j$, is:

$$X_j(\bar{\kappa}_j) \;=\; \frac{I}{J}\int_{\bar{\kappa}_j}^{\infty} \int_{\tilde{\gamma}(k)}^{\infty} dG(\gamma)\, dF(k) \tag{2}$$

which in principle may include patients from both the first and second term of equation (1).

For given formulations of the functions $Q_j(\kappa)$ and $X_j(\kappa)$, each physician solves for the following constrained optimization:

$$\bar{\kappa}_j^* \;\equiv\; \arg\max_{\bar{\kappa}_j} R_j(Q_j) - P(X_j) \qquad \text{s.a.} \quad 0 \leq \bar{\kappa}_j \leq \bar{\kappa}_{\max} \tag{3}$$

where $\bar{\kappa}_{\max} < \infty$ is the maximum value the choice of threshold of any physician may take.

An inner solution to problem (3) takes on the following form:

$$R_j'(Q_j)\frac{\partial Q_j}{\partial \bar{\kappa}_j} = P'(X_j)\frac{\partial X_j}{\partial \bar{\kappa}_j} \tag{4}$$

Proposition 2.1. *In the non-search model as described thus far, when the solution $\bar{\kappa}_j^*$ of problem (3) $\in (0, \frac{\tau_j}{V_j}]$, this $\bar{\kappa}_j^*$ is described by:*

$$R_j'(Q_j(\bar{\kappa}_j)) = P'(X_j(\bar{\kappa}_j))$$

Proof. For $Q_j(\bar{\kappa}_j)$ and $X_j(\bar{\kappa}_j)$ as defined by (1) and (2), when $\bar{\kappa}_j \in (0, \frac{\tau_j}{V_j}]$:

$$\frac{\partial Q_j}{\partial \bar{\kappa}_j} = -\frac{I}{J} \int_{\tilde{\gamma}(\bar{\kappa}_j)}^{\infty} dG(\gamma)\, f(\bar{\kappa}_j) = \frac{\partial X_j}{\partial \bar{\kappa}_j}$$

such that both derivatives are cancelled in equation (4). As a result, if the optimal $\bar{\kappa}_j^* \in (0, \frac{\tau_j}{V_j}]$, the proposition follows. □

This proposition implies that when for low enough values of κ_i, any new patients that physician j attracts are those she entices through sick leave leniency, meaning that in the vicinity of $\bar{\kappa}_j \in (0, \frac{\tau_j}{V_j}]$ the change in expected patients $\Delta Q_j / \Delta \bar{\kappa}_j$ through a variation in $\bar{\kappa}_j$ is the same as the change in expected sick leaves issued $\Delta X_j / \Delta \bar{\kappa}_j$.

Problem (3) offers three types of solutions: two corner solutions at 0 and $\bar{\kappa}_{\max}$, and an inner solution satisfying (4). The proposition below proves that an inner solution can't hold for $\bar{\kappa}_j \in (\frac{\tau_j}{V_j}, \bar{\kappa}_{\max})$.

Proposition 2.2. *In the non-search model as described thus far, no $\bar{\kappa}_j$ in $(\frac{\tau_j}{V_j}, \bar{\kappa}_{\max})$ can be the solution of problem (3).*

Proof. For $\bar{\kappa}_j \in (\frac{\tau_j}{V_j}, \bar{\kappa}_{\max})$,

$$Q_j(\bar{\kappa}_j) \;=\; \frac{I}{J}\left[\int_{\tau_j/V_j}^{\infty} dF(\kappa)\right] \quad \text{and thus} \quad \frac{\partial Q_j}{\partial \bar{\kappa}_j} \;=\; 0$$

whereas

$$\frac{\partial X_j}{\partial \bar{\kappa}_j} = -\frac{I}{J} \int_{\tilde{\gamma}(\bar{\kappa}_j)}^{\infty} dG(\gamma)\, f(\bar{\kappa}_j) \quad < \quad 0$$

so equation (4) can never hold, which would be required of $\bar{\kappa}_j$ as an inner solution to problem (3). □

The intuition is that patients with $\kappa_i \geq \tau_j / V_j$ visit their assigned physician whether or not they get sick leave: they form a "captured clientele". A threshold above τ_j / V_j grants leave only to captured patients, so it attracts no one and only raises her expected fine; she does better by raising it all the way to $\bar{\kappa}_{\max}$. In the search models, patients choose among physicians, so these patients are no longer fully captured: a stricter threshold makes them less likely to choose her, though it never drives that probability to zero.

Equilibrium in the non-search model. *Given patients characterized by $\{(\kappa_i, \gamma_i)\}_{i=1}^{I}$, physicians $\{(V_j, R_j(\cdot), \tau_j)\}_{j=1}^{J}$, and a punishment function $P(\cdot)$, an equilibrium is a set of physician thresholds $\{\bar{\kappa}_j\}_{j=1}^{J}$ that solve problem (3), together with the expected patient demand and sick leaves $\{(Q_j, X_j)\}_{j=1}^{J}$ given by (1) and (2).*

This benchmark, which follows Schnell (2026), already shows features that carry over to the search models: thresholds that rise with physician quality, and possible corner solutions at 0 and $\bar{\kappa}_{\max}$. What it leaves out is patients' choice of physician, making each of them a local monopolist. Letting patients search makes each physician's demand depend on every other physician's threshold, so that thresholds must be determined jointly, in a Nash equilibrium.

2.3 General Search Framework

I introduce patient search as a general framework in which patients are more likely to match with physicians which provide them more utility.

Let $\mathcal{J} = \{1, \ldots, J\}$ denote the set of physicians, and let $j = 0$ denote the outside option of visiting no physician. For each patient i, I define a vector $S_i = (s_{i0}, s_{i1}, \ldots, s_{iJ}) \in \Delta(\mathcal{J} \cup \{0\})$, where s_{ij} is the probability that patient i visits physician j and $s_{i0} = 1 - \sum_{j=1}^{J} s_{ij}$ is the probability that he visits none. Both specifications of S_i that I consider below fold the second and third stages of the game into one: patients set $s_{ij} > 0$ only for physicians they are willing to visit, and $s_{i0} = 1$ if no physician gives them positive utility.

I restrict how s_{ij} depends on the utility patient i expects from each physician. Writing $u_{i,-j}$ for the utilities he expects from the $J - 1$ physicians other than j,

$$s_{ij} \equiv g_i(u_{ij}, u_{i,-j}),$$

where $g_i(\cdot)$ is continuous, weakly increasing in u_{ij} and weakly decreasing in each component of $u_{i,-j}$. The two models will give g_i a specific form.

Defining s_{ij} in terms of u_{ij} makes it depend on patient i's characteristics (κ_i, γ_i). I will use the $s_{ij}(\kappa_i, \gamma_i)$ notation to define physician j's expected demand and sick leaves issued in terms of patients' visit probabilities:

$$Q_j(\bar{\kappa}_j, \bar{\kappa}_{-j}) = I \int_0^\infty \int_0^\infty s_{ij}(\kappa, \gamma)\, dG(\gamma)\, dF(\kappa) \tag{5}$$

$$X_j(\bar{\kappa}_j, \bar{\kappa}_{-j}) = I \int_{\bar{\kappa}_j}^\infty \int_0^\infty s_{ij}(\kappa, \gamma)\, dG(\gamma)\, dF(\kappa) \tag{6}$$

These formulations are quite general. Note that unlike (1) and (2), they depend not only on the physician's own $\bar{\kappa}_j$ but also the thresholds of the other $J - 1$ physicians: $\bar{\kappa}_{-j}$. The final set of thresholds $\{\bar{\kappa}_j^*\}_{j=1}^J$ is a Nash equilibrium between the physicians, with the counterveiling forces of stealing business from others through leniency and the larger risk of a fine that comes from it.

Before moving on to my first model specification, I note that this general formulation of s_{ij} is sufficient to make the next proposition.

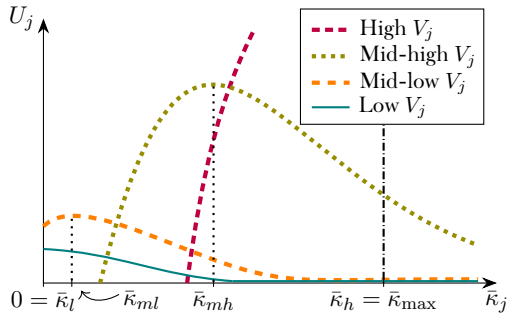
Proposition 2.3. *In the patient search framework as described, and assuming $\gamma_i \geq 0$ for all patients, for two physicians j, l, such that $\bar{\kappa}_j \neq \bar{\kappa}_l$, $\bar{\kappa}_j$ and $\bar{\kappa}_l$ are strategic complements.*

Proof. The proof is in the Appendix A.1. □

2.4 The Explicit Search Model

In the *explicit* search model, which I also call the sequential model, S_i results from sequential search as in McCall (1970), with a reservation utility in place of a reservation wage.

Figure 1: Physician utility in the explicit search model



Notes: Stylized utility curves U_j of four physicians of different quality V_j, as a function of their own threshold $\bar{\kappa}_j$. Vertical lines mark each physician's optimal threshold: the low and high types choose the corners 0 and $\bar{\kappa}_{\max}$, the two middle types an interior threshold.

Each period patient i draws one physician at random, with replacement and without recall, and either visits her, receiving u_{ij} in every period thereafter, or draws again next period, discounting the future at $\beta \in (0, 1)$.

This is an optimal stopping problem, and its solution is a reservation rule: the patient visits the first physician he draws who gives him at least his reservation utility $\bar{U}_i$, which depends on the quality, leniency and visit costs of all physicians. I show in Appendix A.2 that $\bar{U}_i$ solves:

$$\bar{U}_i = \frac{\beta}{1-\beta} \frac{\sum_{j=1}^{J} \mathbb{1}[u_{ij} \geq \bar{U}_i] \cdot (u_{ij} - \bar{U}_i)}{\sum_{j=1}^{J} \mathbb{1}[u_{ij} \geq \bar{U}_i]} \tag{7}$$

The visit probabilities s_{ij} in each patient's S_i are then defined in terms of patient i's reservation utility $\bar{U}_i$:

$$s_{ij} = \frac{\mathbb{1}[u_{ij} \geq \bar{U}_i]}{\sum_{k=1}^{J} \mathbb{1}[u_{ik} \geq \bar{U}_i]} \tag{8}$$

Equation (8) makes the simplifying but reasonable assumption that patient i draws physicians randomly and thus has equal probability to visit any of the physicians who give him utility above his threshold $\bar{U}_i$.

The parameter β has its usual interpretation of *patience*, but in doing so it governs the dispersion of patient demand. As it approaches 1, patients are more willing to wait until they get a good draw, and so patient demand becomes more concentrated on the high quality physicians. Lowering β is akin to introducing noise to patient search.

Figure 1 is a stylized representation of the utility of four "types" of physician, ordered by their level of service quality V_j, as a function of their own threshold $\bar{\kappa}_j$. The 'low' and 'high' types are driven to a corner solution at 0 and $\bar{\kappa}_{\max}$, respectively, while the two 'middle' types choose inner solutions, which grow with V_j.

Equilibrium for the sequential model is formalized below.

Equilibrium in the "explicit" search model. *Given patients characterized by* $\{(\kappa_i, \gamma_i)\}_{i=1}^I$*, physicians* $\{(V_j, R_j(\cdot), \tau_j)\}_{j=1}^J$*, and search parameter* β*, an equilibrium is a set of patient thresholds* $\{\bar{U}_i\}_{i=1}^I$ *satisfying (7), patient visit probabilities* $\{S_i\}_{i=1}^I$*, with* $S_i = (s_{i1}, \ldots, s_{iJ})$ *given by (8), and physician thresholds* $\{\bar{\kappa}_j\}_{j=1}^J$ *that solve problem (3), together with the expected patient demand and sick leaves* $\{(Q_j, X_j)\}_{j=1}^J$ *given by (5) and (6).*

2.5 The Implicit Search Model

The *implicit* search model, which I also call the logit model, replaces the sequential search process with a logit discrete choice model, which approximates the explicit search model while being simpler to compute. I modify the logit so that patients only choose among the physicians who give them positive utility, as free disposal requires.

I call it "implicit" search because visit probabilities are not micro-founded in a model of patient search like the previous one, instead taking the form of a standard reduced form discrete choice, yet arriving at qualitatively similar distributions of patient demand over physician quality and leniency.

I define the visit probabilities s_{ij} in the patient's S_i as follows:

$$s_{ij} = \frac{\alpha_{ij}}{\sum_{k=1}^J \alpha_{ik}}, \quad \text{where} \quad \alpha_{ij} = \begin{cases} e^{\lambda u_{ij}}, & \text{if} \quad u_{ij} > 0 \\ 0, & \text{if} \quad u_{ij} \leq 0 \end{cases} \qquad \text{and} \quad \lambda \in \mathbb{R}^+ \tag{9}$$

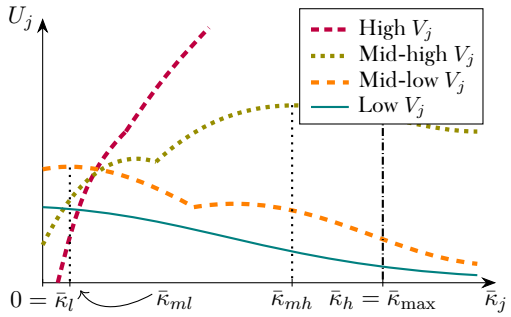
The parameter λ plays a similar role as β did in the sequential model: the lower its value, the noisier patient "search" is, giving higher match probability to suboptimal but net-positive physician choices. On the other hand, as $\lambda \to \infty$, patient choice is concentrated with probability approaching 1 at the physician with the highest V_j.

Unlike the explicit search model, the probability that a given patient i visits physician j is *strictly* growing in κ_i, rather than being a piecewise constant function with different levels. There is a discrete jump in probability at $\bar{\kappa}_j$, but elsewhere above 0 the function is smoothly increasing in physician quality V_j.

Figure 2 shows physicians at the same four levels of quality as Figure 1. The 'low' and 'high' ones are again driven to the corners, but the curves in between are no longer single-peaked: they look as though composed of two concave curves joined together. The reason is that for each physician j there is a level $\kappa_i = \tau_j / V_j$, as in the non-search model, above which patients always give positive probability to visiting her. Granting sick leave to those patients raises their s_{ij}, but taking it away doesn't drive it to 0, as it does for patients with $\kappa_i < \tau_j / V_j$.

A raise in $\bar{\kappa}_j$ therefore loses patients at a different rate on the $[0, \tau_j/V_j]$ and $[\tau_j/V_j, \bar{\kappa}_{\max}]$ sections of the curve, and for some values of V_j each section has its own peak. A physician's optimal threshold is at whichever peak is higher, if not in a corner solution. I call 'mid-low' the physicians whose first peak is higher than the second, and 'mid-high' the others, as they have a higher V_j in the parametrizations I use later in this paper.

Figure 2: Physician utility in the implicit search model



Notes: Stylized utility curves U_j of four physicians of different quality V_j, as a function of their own threshold $\bar{\kappa}_j$. Vertical lines mark each physician's optimal threshold. The two middle types have two local maxima and are told apart by which of the two is higher.

The breaking point of the curve occurs earlier the higher the V_j because physicians of higher quality do not charge proportionally higher fees τ_j in my simulations in Section 3.2, on which I based this graph, but this need not be the case. Notice also that $\bar{\kappa}_j$ is growing in V_j in this and the previous figure, which isn't an analytical property of the models, but a result of using continuous, unimodal distributions for $F(\kappa)$ and $G(\gamma)$.

As before, I formalize the notion of equilibrium in the implicit search model.

Equilibrium in the "implicit" search model. *Given patients characterized by* $\{(\kappa_i, \gamma_i)\}_{i=1}^{I}$*, physicians* $\{(V_j, R_j(\cdot), \tau_j)\}_{j=1}^{J}$*, a punishment function* $P(\cdot)$*, and search parameter* λ*, an equilibrium is a set of patient visit probabilities* $\{S_i\}_{i=1}^{I}$*, with* $S_i = (s_{i1}, \ldots, s_{iJ})$ *given by (9), and physician thresholds* $\{\bar{\kappa}_j\}_{j=1}^{J}$ *that solve problem (3), together with the expected patient demand and sick leaves* $\{(Q_j, X_j)\}_{j=1}^{J}$ *given by (5) and (6).*

3 Computation

3.1 Algorithms for Model Computation

Given a set of physicians $\{(V_j, R_j(\cdot), \tau_j)\}_{j=1}^{J}$, punishment function $P(\cdot)$, distribution functions for patient parameters $F(\kappa)$ and $G(\gamma)$, and one search model parameter z, which is λ in the logit model, β in the sequential model, I develop an algorithm to compute market equilibria.

Equilibrium aggregates are computed by Monte Carlo simulation, using vectorized matrix operations. Set J as the number of physicians,[7] $\tilde{I}$ as the size of the sample drawn randomly from $F(\kappa)$ and $G(\gamma)$. For both models I define a class which can output a matrix S where each column is a patient's visit probability vector S_i when given as input the arrayed set of

[7]Or, as I'll later interpret it, as the number of bins, where each bin j is a unique combination of $(V_j, \tau_j, R_j(\cdot))$.

physicians' quality and visit cost $\{(V_j, \tau_j)\}_{j=1}^{J}$, an arrayed set of patients $\{(\kappa_i, \gamma_i)\}_{i=1}^{\tilde{I}}$, the model parameter z and a given vector of physician strategies $\{\bar{\kappa}_j\}_{j=1}^{J}$.

I input as "patients" my $\tilde{I}$ samples from $F(\kappa)$ and $G(\gamma)$, then a $J \times \tilde{I}$ matrix U is computed, where each component u_{ji} corresponds to the utility the sampled patient i would get from a visit to physician j.[8] This step is the same for both classes, i.e. for both models.

What differs between the two models is the computation of the matrix S of patient visit probabilities out of the utility matrix U:

- **Implicit search model (logit):** First, an 'α-matrix' is calculated over matrix U, where each α_{ji} is $e^{\lambda u_{ji}}$ if $u_{ji} > 0$ and 0 if not. Then, for each patient i, that is, for each column, each component s_{ji} of the S matrix takes on the values $s_{ji} = \alpha_{ji} / \sum_{k=1}^{J} \alpha_{ki}$.

- **Explicit search model (sequential):** Recall Equation (7) characterizing patient thresholds. I first compute the $\tilde{I}$-dimensional vector of the sampled patients' respective $\bar{U}_i$. Define U_i as the set $\{u_{ji}\}_{j=1}^{J}$ of utilities patient i receives from a visit to each physician. In matrix terms, U_i would be the ith column of the $J \times \tilde{I}$ matrix U. The operation to compute each $\bar{U}_i$ is the following:

$$\bar{U}_i \equiv \underset{x \in U_i}{\arg\min} \left\| x - \frac{\beta}{1-\beta} \frac{\sum_{j=1}^{J} \mathbb{1}[u_{ji} \geq x] \cdot (u_{ji} - x)}{\sum_{j=1}^{J} \mathbb{1}[u_{ji} \geq x]} \right\|$$

  where the norm $\| \cdot \|$ is defined in $\mathbb{R}$ as simply the absolute value $| \cdot |$. This is to say, for each sampled patient i I evaluate x for each u_{ji} in U_i. Each $u_{ji} \in U_i$ is thus a candidate reservation utility, corresponding to the rule "visit only physicians at least as good as j", and $\bar{U}_i$ is the candidate that best satisfies (7). This is computationally less intensive than seeking to compute the exact root of Equation (7), which would be a redundant exercise, because there's a discrete number of physicians above that mark, and selecting to use "the lowest u_{ji} above the root of (7)" as threshold instead of the exact root would result in the same vector of strategies S_i for each patient.[9]

  Once having computed the $\tilde{I}$-dimensional vector of all sampled patients' $\bar{U}_i$, I evaluate column-wise the indicator function $\mathbb{1}[u_{ji} \geq \bar{U}_i]$ over matrix U, and the $J \times \tilde{I}$ matrix S of patient visit probabilities takes on the values $s_{ji} = \mathbb{1}[u_{ji} \geq \bar{U}_i] / \sum_{k=1}^{J} \mathbb{1}[u_{ki} \geq \bar{U}_i]$.

The steps following the computation of the S matrix are the same for both models. The J-dimensional vector Q of each physician's expected demand Q_j is achieved through the row-wise summation of all patients' demand for j, i.e. performing $\sum_{i=1}^{\tilde{I}} s_{ji}$ for each j. The vector X of each physician's expected sick leaves issued X_j performs that same summation, conditional on the patients' κ_i being above or at physician j's chosen threshold $\bar{\kappa}_j$, that is, $\sum_{i=1}^{\tilde{I}} \mathbb{1}[\kappa_i \geq \bar{\kappa}_j] \, s_{ji}$. Both sums are normalized to fit the actual number of patients in the physician-patient market, multiplying them by $I/\tilde{I}$.

[8] As I have defined my matrices $J \times \tilde{I}$ for ease of visualization, I will refer to matrix components as x_{ji} in this section rather than x_{ij} as I do elsewhere in the paper.

[9] Granted, this is not strictly true, as in my formulation a 'u_{ji}' may be selected as threshold which is actually below the exact root in $\mathbb{R}$, but closer to it than the nearest one above it. The effect of this "estimation noise" on my overall results is negligible.

For a given vector $\bar{\kappa}$, computing patient visit probabilities, physician aggregates and then physicians' utility (defined as $\{R_j(Q_j) - P(X_j)\}$) entails a sequence of calculations done in linear time. However, the calculation of the *equilibrium* vector $\bar{\kappa}^*$ is performed in quadratic time $\mathcal{O}(n^2)$, as it requires each of the aforementioned steps to be done some x number of times *for each* physician, such that the inclusion of an additional physician increases the cost of each evaluation as well as the amount of physicians whose $\bar{\kappa}_j$ needs to be computed.

The algorithm to find the equilibrium J-vector $\bar{\kappa}^*$ of physician strategies is as follows:

i Input an initial guess $\bar{\kappa}^0$ for physician thresholds.

ii For each physician, I fix the value of $\bar{\kappa}_{-j}$, the threshold values of the other $J-1$ physicians, and compute the aggregates Q_j and X_j and physician j's utility for different choices of $\bar{\kappa}_j$ across a grid. In particular, $U_j(\bar{\kappa}_j, \bar{\kappa}_{-j})$ is computed for each decimal step between 0 and $\bar{\kappa}_{\max}$, the maximum threshold physicians may choose.

iii The choice of threshold k_j which gave j the most utility in the previous step is selected, and a grid is set up spanning the 19 values in $[k_j - 0.09, k_j + 0.09]$ spaced by hundredths,[10] each value therein input as physician j's threshold $\bar{\kappa}_j$ to compute U_j again, and the value within the grid which maximizes utility is chosen as $\bar{\kappa}_j^1$.

iv Having performed the previous step for all J physicians, I input as a new guess the vector $\bar{\kappa}^1 = \{\bar{\kappa}_1^1, ..., \bar{\kappa}_J^1\}$ to run steps ii and iii again. This defines an equilibrium-searching loop $\bar{\kappa}^n = \Phi(\bar{\kappa}^{n-1})$, and the loop is concluded when a fixed point is found, that is, the vector $\bar{\kappa}^*$ such that $\bar{\kappa}^* = \Phi(\bar{\kappa}^*)$.

v Optionally, having found a two-decimal fixed point $\bar{\kappa}^*$, the algorithm can refine it to an x-decimal fixed point $\bar{\kappa}^{*x}$. It starts by running a modified version of step iii on the two-decimal solution, setting up a grid of the 19 values in $[\bar{\kappa}_j^* - 0.009, \bar{\kappa}_j^* + 0.009]$ spaced by *thousandths* for each physician j,[11] selecting that which gives the highest utility for each, and inputting this new vector as guess to run this step again, and so on until the fixed point $\bar{\kappa}^{*3} = \Phi(\bar{\kappa}^{*3})$ is found.

The same procedure then computes the t-decimal solution from the $(t-1)$-decimal solution by setting up grids of step size 10^{-t} for each physician in each iteration, until the solution vector $\bar{\kappa}^{*x}$ reaches the desired number of decimals.

Once the equilibrium vector $\bar{\kappa}^*$ is calculated, it may be used to compute the equilibrium aggregates Q_j and X_j for all physicians, as well as physician utility.

3.2 Illustrative Examples

I use the following parametrization for illustration purposes:

$F(\kappa)$	=	$U[0,1]$	$R(x)$	=	$1.5x$
$G(\gamma)$	=	$U[0,1]$	$P(x)$	=	$\frac{1}{200}x^2$
V_j	$\sim$	$U[0,10]$	τ	=	1

[10] If k_j is 0 (or $\bar{\kappa}_{\max}$), only k_j and the 9 hundredths above (below) it are computed.

[11] For $\bar{\kappa}_j^* = 0$ or $\bar{\kappa}_{\max}$ a similar logic to the footnote above follows.

In particular, I consider $J = 50$ physicians whose V_j is evenly spaced in increments of 0.2 between 0.2 and 10. They share the same $R_j(\cdot)$ and τ_j.

Figure 3 shows equilibrium outcomes in both models under this parametrization: the sequential model for three values of β in the left column, and the logit model for three values of λ in the right column. The figures also show a non-search benchmark where patients are randomized across physicians as in Section 2.2, in which the distribution of patient demand is much flatter and physicians settle at maximum strictness earlier in the quality distribution.

Consider first the sequential model. Something to note is that β is restricted to values within $(0, 1)$. As $\beta \to 1$, Q_j behaves similarly to when $\lambda \to \infty$ in the logit model, which is why the concentration of patients at high V_j's with $\beta = 0.95$ is much more pronounced in Panel 3a than it is with $\lambda = 1$ in Panel 3d.

The shape of the demand curve points to two disimilarities from the logit model:

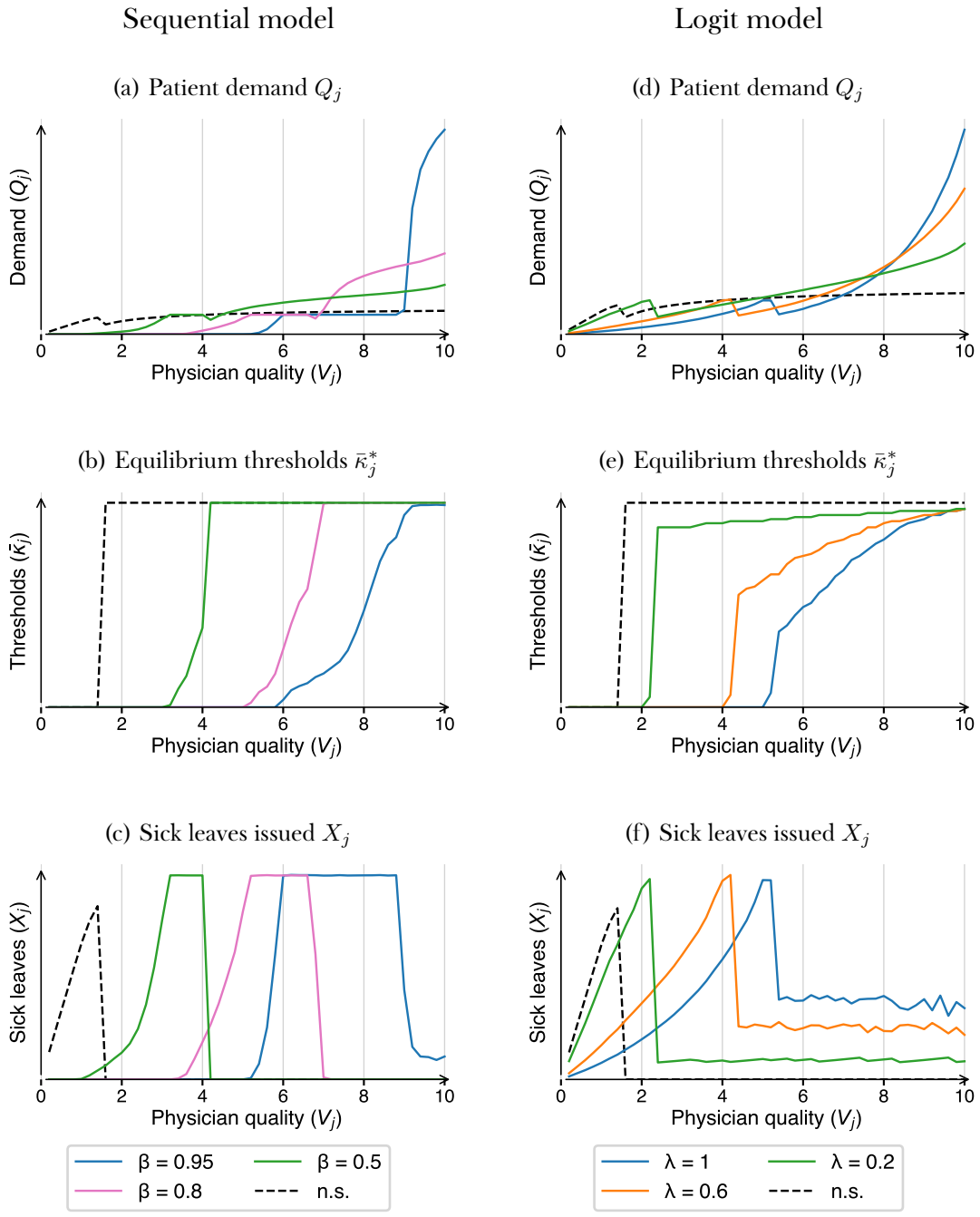
- For physicians of low enough V_j demand is strictly 0 rather than just low, as they don't afford to any patient utility over their threshold $\bar{U}_i$.
- At high values of V_j the demand curve is concave rather than convex as in Panel 3d. This is because the formulation of the 'explicit' search model doesn't continuously (and exponentially) reward higher values of V_j, it only registers whether V_j is enough to have u_{ij} over patients' reservation utility $\bar{U}_i$. Past a certain point, most patients' $\bar{U}_i$ lay below the u_{ij} a certain physician j affords, and incremental gains to even higher V_j are low, hence the concavity.

Panels 3b displays the vector $\bar{\kappa}^*$ of physicians' equilibrium strategies for different values of the model parameter β. To understand the curves' behavior it is useful to refer back to equation 1 for the non-search benchmark. It showed how physicians' expected demand is composed of two types of patients: those who would visit her regardless of leniency due to her quality V_j, and those visiting conditional on $\bar{\kappa}_j$. Physicians of lower quality V_j are incentivized to be lenient to attract clientele: at the lowest end, they are as lenient as possible (in equilibrium), and at a certain point they begin raising their strictness threshold to manage demand, until their quality is enough that they maximize their utility solely from the first group of patients, so they raise their threshold $\bar{\kappa}_j$ to the maximum possible level. The shift from leniency to strictness happens at a higher V_j the higher β is, as demand concentrates in higher-quality physicians and leniency remains necessary for physicians further up the quality distribution.

Equilibrium thresholds in the logit model (Panel 3e) behave similarly, except that they never reach the maximum: since the logit formulation keeps rewarding higher quality, physicians raise their threshold gradually with V_j instead of settling at $\bar{\kappa}_{\max}$.

Sick leaves issued in Panels 3c and 3f mirror the equilibrium strategies. In the sequential model, physicians of low enough quality have no expected demand and so issue none; issuance then rises slowly with quality as lenient physicians gain a clientele, only to drop back to zero past the tipping point of maximum strictness (for illustration, $\bar{\kappa}_{\max}$ is 1, which excludes all patient κ's). The logit model behaves similarly, but past the tipping point high-quality physicians settle at a positive level of issuance, higher the larger λ is, i.e. the more effective patient search is. The logit model continuously rewards leniency.

Figure 3: Equilibrium outcomes in the two search models



Notes: Each panel plots an equilibrium outcome against physician quality V_j, under the parametrization of Section 3.2 with $J = 50$ physicians. The left column is the explicit (sequential) search model for three values of β, the right column the implicit (logit) search model for three values of λ. The dashed line (n.s.) is the non-search equilibrium under the same parameters.

Table 1: Sick leaves per physician and per patient

Statistic	Physicians	Patients
Mean	88	3
S.D.	199	3
Min	1	1
Max	8,743	73
Percentiles		
10th	2	1
50th	24	2
90th	216	7
99th	1,007	14

Notes: Summary statistics of the number of authorized sick leaves issued by each physician and received by each patient between January 2018 and October 2022. The sample contains 4,289,479 sick leaves, 48,611 physicians and 1,445,696 patients.

4 Data

I now turn to the empirical section of this paper. To estimate my models, I will use administrative data provided by FONASA from all sick leaves issued from January 2018 to October 2022 in the public health system. Each observation consists in a sick leave, its status, encrypted ID's for the physician and patient, duration, and *cie10* classification of the condition. I consider authorized sick leaves only. My sample is composed of 4,289,479 observations, including 48,611 different physicians and 1,445,696 different patients.

Table 1 reports descriptive statistics on sick leaves per physician (X_j) and per patient (X_i). Both distributions are strongly right-skewed. The median physician issued 24 sick leaves over the sample period, but the mean is 88. The tail is long: the 90th percentile physician issued 216, the 99th issued 1,007, and the top physician issued 8,743, more than eight times the 99th percentile.

The pattern is similar for patients. Many are only issued sick leave once in the dataset, and the median patient received just two over the sample period. The 90th and 99th percentiles tally 7 and 14, and the patient with the most sick leaves was issued 73, more than five times the 99th percentile.

Long tails are not only present in both distributions, but interrelated. Oteíza (2024) argues that there is a meaningful degree of assortative matching in this sample, where physicians willing to grant many sick leaves are matched with, or found by, patients with a high demand for them. My models capture this phenomenon, as physicians of lower quality endogenously choose to attract patients of low medical need κ_i and high tase for sick leave γ_i through leniency.

The *cie10* classification of each leave might seem a natural source of information on κ, but I do not use it to estimate its distribution, for two reasons. First, κ_i is a willingness to pay for a leave, which need not map one-to-one into medical severity. Second, and more importantly, diagnoses are recorded by the very physicians whose leniency this paper

studies, who might misreport or exaggerate patient conditions. The concern is sharpest for the most common categories in the data—back pain, depression, anxiety, and stress (Table 3, Appendix A.3)—which rest largely on self-reported symptoms and are hard to verify even for a well-intentioned physician, who would have little incentive to question their patients (Carlsen et al., 2020). This is one reason I estimate a structural model, which treats κ as latent and recovers its distribution from observed choices, rather than reading patient types and physician leniency off reported diagnoses.

5 Estimation

5.1 Data Moments

I now turn to calibrating my *implicit* search model using that data. My main challenge is the fact that I have no information on overall patient visits by physician (Q_j), only the sick leaves they were authorized to issue in the sampled period (X_j). I get the necessary variation to be able to identify and estimate my parameters from two sources: by assigning each of the 48,611 physicians to one of ten equally sized quality bins (or decile ranks) and by making a distinction between overall sick leaves granted and those of 30 days or more.

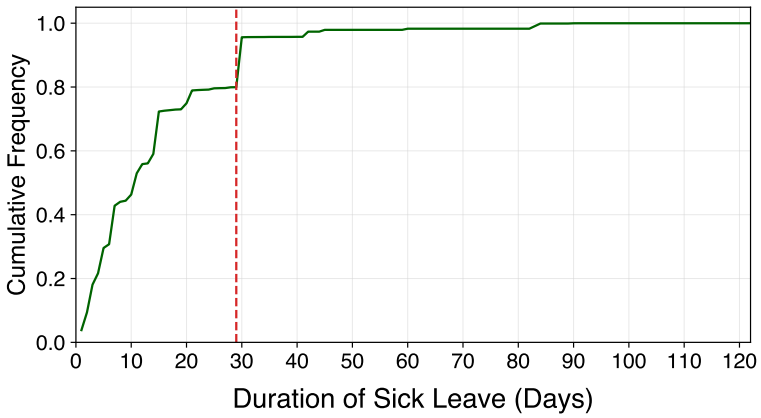
A qualitative consequence of both my proposed models is that physician strategies $\bar{\kappa}_j$ (strictness) were growing on physician quality V_j, which for my choice of distributions for $F(\kappa)$ and $G(\gamma)$ holds monotonically. Lacking data on observable qualities from physicians, including individual salary and educational background, I group them into quality bins based on properties derived from the models: higher quality physicians will raise their threshold $\bar{\kappa}_j$ and thus see patients of higher κ_i on average, and in particular, a higher proportion of their patient base will be composed of patients satisfying $\kappa_i > \bar{\kappa}_{\max}$.

Here I can use the intensive margin in my data: I observe each sick leave's *duration* (in days). I argue that the empirical counterpart of patients with $\kappa_i > \bar{\kappa}_{\max}$ is those who received sick leaves of 30 or more days. The medical severity of a condition requiring 30 days rest or more is considerable, and although there may be variation in the intensive margin of duration, it is implausible that a competent physician, faced with the same diagnosis, would deny leave outright out of strictness, so these are visits that would have resulted in granted sick leave from any physician in my sample. Moreover, 30 days is a threshold beyond which certification faces closer scrutiny and requires external approval. This eases concerns about misreporting and makes it a safe moment to calibrate on.

Figure 4 illustrates the empirical cumulative distribution of sick leave durations across all observations in my dataset.[12] Sick leaves lasting 30 days or more account for 20.05% of issuances during the sample period. In particular, sick leaves of exactly 30 days represent 15.64% of the total, a significant discrete jump in the curve which afterwards becomes almost flat. The highest duration of any single sick leave in my sample is 728 days.

[12]Since this probability distribution assumes integer values only, the cumulative probability of a sick leave lasting 30 or more days is calculated as $1 - \text{CDF}(29)$.

Figure 4: Cumulative distribution of sick leave durations



Notes: Empirical cumulative distribution of the duration, in days, of all sick leaves in the sample. The dashed red line marks 29 days; sick leaves of 30 days or more account for 20.05% of the total. The horizontal axis is cut at 120 days, while the longest sick leave lasts 728 days.

I first make use of this distinction to set up my physician quality bins. I design a *quality score* variable by physician based on three criteria:

i Average duration of sick leaves issued.

ii Percentage of sick leaves that are 30+ days in sampled period.

iii Amount of sick leaves of 30+ days issued.

The quality bins correspond to the decile ranks of the quality score. See Appendix A.3 for a breakdown of the construction of the score variable and the physician bins.

There are admittedly some concerns of circularity in using predictions of the model to classify physician quality, but note that the sample moments with which I calibrate my parameters contain, in part, different information from that used to set up the bins themselves, so that identification does not rest on the sorting alone. The purpose of this exercise is to evaluate policy assuming my models capture key features of the setting, not to test those features against the data.

The distinction between total sick leaves and those of 30 or more days also serves my calibration targets: for each quality bin I match both, giving 20 moments in all. Variation comes only from the bins, as r_j and τ_j are equalized across physicians.

I treat each quality bin as a single representative physician, so that $j = 1, \ldots, J$ indexes bins, with $J = 10$. The model counterparts of the two moments for bin j are

$$X_j = I \int_{\bar{\kappa}_j}^{\infty} \int_0^{\infty} s_{ij}(\kappa, \gamma)\, dG(\gamma)\, dF(\kappa), \tag{6}$$

$$X_j^{30} = I \int_{\bar{\kappa}_{\max}}^{\infty} \int_0^{\infty} s_{ij}(\kappa, \gamma)\, dG(\gamma)\, dF(\kappa). \tag{10}$$

These are *expected* sick leaves in a population of I patients: since s_{ij} is the probability that patient i visits physician j, they average over the outcomes of patients' search. Their sample counterparts, $\hat{X}_j$ and $\hat{X}_j^{30}$, are the sick leaves *actually issued*, in total and of 30 or more days, by the physicians in bin j during the sample period. As I has no analogue in the data, I express all moments as shares of total sick leaves, matching $X_j / \sum_k X_k$ to $\hat{X}_j / \sum_k \hat{X}_k$, and likewise for X_j^{30}. Stacking across bins gives the J-dimensional vectors X, X^{30}, $\hat{X}$, and $\hat{X}^{30}$.

The advantage of using this second defined moment, X^{30}, lies in its independence from the choice of strategy $\bar{\kappa}_j$ and its looser dependence from the distribution $G(\gamma)$. Any patient satisfying $\kappa_i > \bar{\kappa}_{\max}$ will, by definition, be granted sick leave by any physician. The implication of this on the probability s_{ij} that such a patient visits any physician j can be seen as follows:

$$s_{ij} = \frac{\alpha_{ij}}{\sum_{k=1}^{J} \alpha_{ik}} = \frac{e^{\lambda u_{ij}}}{\sum\limits_{k:\ u_{ik} \geq 0} e^{\lambda u_{ik}}} = \frac{e^{\lambda\{V_j \kappa_i + \gamma_i - \tau_j\}}}{\sum\limits_{k:\ u_{ik} \geq 0} e^{\lambda\{V_k \kappa_i + \gamma_i - \tau_k\}}} = \frac{e^{\lambda\{V_j \kappa_i - \tau_j\}}}{\sum\limits_{k:\ u_{ik} \geq 0} e^{\lambda\{V_k \kappa_i - \tau_k\}}}$$

Since γ_i enters every exponent additively, it cancels from the fraction. It still matters, however, through the summation in the denominator: by free disposal, patient i visits physician j with positive probability only if $u_{ij} \geq 0$, that is, $V_j \kappa_i + \gamma_i \geq \tau_j$. The scale of γ_i thus affects X^{30} only insofar as it removes physicians from a patient's choice set altogether. The identification panels in the Appendix of Tagle (2024) show that the X^{30} moments respond much less than the X moments to the parameters governing $G(\gamma)$.

5.2 Parameters and Estimation Results

There are several elements in this model that require specification and then calibration. Starting with the two population distributions describing patients, $F(\kappa)$ and $G(\gamma)$, and in keeping with their analogues in Schnell (2026), I choose an exponential distribution for κ with rate parameter λ_F, so that higher levels of medical necessity (*pain severity* in Schnell) become increasingly rare. For γ, the taste for sick leave (the taste for prescription drugs in Schnell), I choose a normal distribution of mean μ and standard deviation σ.[13]

Both μ and σ I will fit through calibration, but for λ_F I simply normalize it to 2—such that the population average κ_i is $1/2$—because I don't have a cardinal benchmark for the value of my quality bins $V_1 ... V_{10}$ which scale the effect of κ_i in the patient utility function. By fixing λ_F, I can calibrate the value of the quality bins. This is innocuous: changing λ_F to k^* rescales every κ_i by $2/k^*$, which is undone by scaling each V_j by $k^*/2$ and $\bar{\kappa}_{\max}$ by $2/k^*$, leaving the equilibrium unchanged. I could have alternatively normalized V_1 to 1.

For physicians, I estimate the ten components of the $\vec{V}$ vector. In their utility function, I take the revenue function $R_j(\cdot)$ to be linear, $r_j Q_j$, and the punishment function $P(\cdot)$ to be strictly convex and quadratic, $\frac{p}{2} X_j^2$. For convenience, I drop the j subscript on revenue, as I have done all along for $P(\cdot)$. I estimate p by SMM. The revenue per visit r and the cost of a visit τ, which I also take to be common across physicians, are taken from the data directly as weighted averages of FONASA's listed prices for doctor visits, matched to the ten conditions most often granted sick leave (see Appendix A.3).

[13] This violates the assumption of $\gamma_i \geq 0$, but the calibrated negative mass is trivial: 0.003% of patients.

Table 2: Model parameters

Parameter	Value	Source
Physician quality bins		
V_1 ... V_{10}	$\vec{V}_{1\times 10}$	SMM
Punishment function $P(\cdot)$		
p	1.953	SMM
Medical need distribution $F(\kappa)$		
λ_F	2	Normalization
Taste distribution $G(\gamma)$		
μ	73.997	SMM
σ	18.492	SMM
Logit model parameter		
λ_s	0.842	SMM
Maximum threshold level		
$\bar{\kappa}_{\max}$	0.763	SMM
Revenue/cost of visit		
r	21.182	Data
τ	8.944	Data

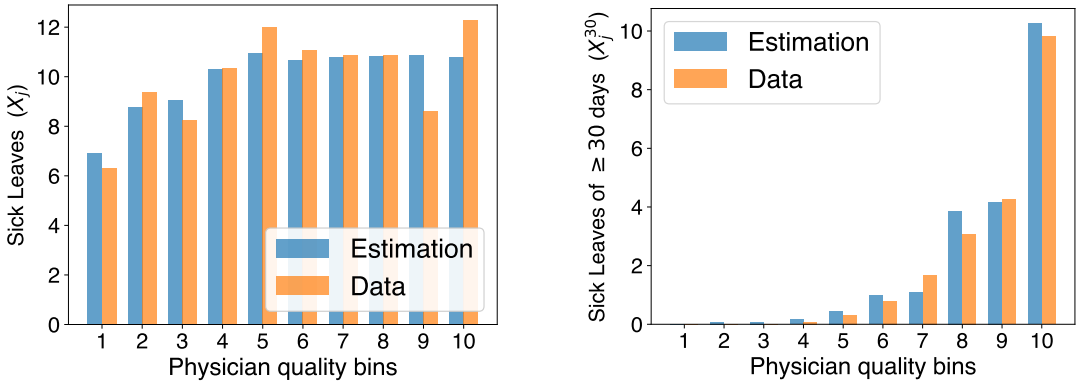
Notes: Parameters marked SMM are estimated by the simulated method of moments; the ten values of V_j are not reported individually. λ_F is normalized. r and τ are computed from FONASA's fee schedule, in thousands of Chilean pesos (Appendix A.3).

The choice of a linear function for $R_j(\cdot)$ is natural, the parameter r has a direct interpretation as physician revenue by visit. The choice of $P(\cdot)$ is a little more subtle. At this point I would like to point out that it is not self-evident that my *punishment* function should take total sick leaves granted by physician (X_j) as its argument. The first-best option would also take into account patients' medical need κ_i, but this is unobservable to the institutional overseer, which relies on physicians' own reporting of the patients' condition.

My setting has a precedent of institutional discipline relying on observable aggregates. As studied by Oteíza (2024), in 2021 Chile's regulatory authority sanctioned a subset of physicians who were outliers in their volume of sick leaves issued and were unable to justify it. Let f be the amount associated with a fine for over-issuance, and the expected punishment function could be exressed as $f \times \Pr[f \mid X_j]$. Empirically, $\Pr[f \mid X_j]$ would be zero until around the 90th percentile of issuance level X_j, after which the probability rises steeply. I approximate this convex function with the tractable quadratic $\frac{p}{2}X_j^2$, which is less steep than the true function but more parsimonious, such that p captures both the size and the probability of a fine.

Finally, I estimate two more parameters: λ_s, specific to the logit model, which governs how effective search is, and $\bar{\kappa}_{\max}$, the upper bound on physician thresholds, which governs

Figure 5: Model fit: moments in the data and in the model



(a) All sick leaves, X_j

(b) Sick leaves of 30+ days, X_j^{30}

Notes: The twenty moments used in estimation, by physician quality bin, in the data (orange) and in the fitted model (blue). Moments are expressed as a percentage of all sick leaves in the sample.

the aggregate level of X^{30}. At its estimated value of 0.763, 21.75% of patients fall above this bound, close to the 20.05% of sick leaves in my sample that last 30 days or more.

The role, value and source of the model parameters are summarized in Table 2. Two parameters are directly retreived from the data (r and τ), one is normalized (λ_F), the rest are subject to SMM estimation. This leaves 15 parameters to estimate from 20 data moments, so the model is overidentified.

Define $\theta = \{V_1, \ldots, V_{10}, p, \mu, \sigma, \lambda_s, \bar{\kappa}_{\max}\}$ and $m = \{X_1, \ldots, X_{10}, X_1^{30}, \ldots, X_{10}^{30}\}$. The simulated method of moments (SMM) defines the estimator as:

$$\hat{\theta}_{SMM} = \arg\min_{\theta} \| \hat{m} - m(\theta) \| \tag{11}$$

where $\hat{m}$ is the vector of moments computed from the data and $m(\theta)$ is the vector of the corresponding moments generated by the model at parameters θ. The estimator $\hat{\theta}_{SMM}$ is the set of parameters that minimizes the distance between the two.

Minimization is achieved through a series of global and local optimization algorithms, using the '*dual_annealing*' and '*Powell*' modules of Scipy optimization.

Table 2 presents the resulting set $\hat{\theta}_{SMM}$ of estimated parameters. Figure 5 shows the comparison between the moments in the data $m(x)$ and the moments from my SMM-fitted model $m(x|\hat{\theta}_{SMM})$ as bar plots.

The first four representative physicians take a corner, $\kappa_j = 0$ solution. The rest are at the inner solution, satisfying the FOC equation (4), choosing an increasingly large threshold in relation to their higher V_j. Patients satisfying $\kappa_i > \bar{\kappa}_{\max}$, who place high value on physician quality, are concentrated at the highest bins, particularly in bin 10.

5.3 Identification

Though satisfying the minimal condition that I have more moments (20) than equations (15), I need to further ensure the correct identification of each individual parameter. This is straightforward to argue for in the case of the $\vec{V}$ vector, each of the ten V_j parameters directly raises demand for the representative physician j, increasing the value of both moments for the respective quality bin. Identification for the other five moments comes from the interplay between the ten X moments and the other ten X^{30} moments.

The master's thesis version of this paper (Tagle, 2024) includes Appendix figures that plot how the distributions of X_j and X_j^{30} across quality bins change as each of the five remaining parameters varies, as a way of making visual how they are individually identified. Below I briefly explain for each parameter which variation in the data pins it down:

λ_s This parameter determines the "quality" of search in the logit model. As it increases towards ∞, demand concentrates exclusively at V_{10}. As it tends to 0, demand is uniformly distributed across the ten bins. This sensitivity is more pronounced for the X^{30} moments—sick leaves for patients above $\bar{\kappa}_{\max}$—for the reasons mentioned in Section 5.1: this moment is almost independent from γ_i, which conversely makes it more sensitive to V_j, which is in turn scaled by λ_s. This parameter serves to explain, for a given $\vec{V}$, the relative steepness of the distribution of X^{30} with respect to X.

$\bar{\kappa}_{\max}$ This threshold directly controls the scale of overall X^{30} with respect to X. The lower it is, the more aggregate X^{30} approaches aggregate X. It should be such that the X^{30}/X ratio resembles the ratio of sick leaves of 30+ days over total sick leaves issued in the data, as it currently does (21.75% vs. 20.05%).

p The *punishment* function parameter p determines the level of X_j at which physicians who choose an inner solution $\bar{\kappa}_j$ stay: their first-order condition $r = pX_j$ fixes it at r/p, regardless of their V_j. It also determines the first physician bin to play an inner solution. This is why the six highest-quality representative physicians in Figure 5 lie on a horizontal line, and raising or lowering p moves that line down or up.

μ, σ Here identification is only partial. Jointly, they *can* be identified, as they account for variations in the location of the first inner solution bin beyond what the p parameter can achieve, but they can't be readily separated. What matters is the total mass up to a given threshold $\bar{\kappa}_j$, not the shape of the curve, so the effect of a decrease in μ is equivalent to that of an increase in σ. In this regard I can say that μ and σ jointly describe a feasible plane where the distribution I achieved is possible for values in both parameters, but they fail at reducing this 2-dimensional plane to a singleton, which would be the exact identification of both.

I could have also normalized σ to a single value or a ratio relative to μ, without loss. For now, I go on with my estimated parameters and run simulations to showcase my framework.

6 Counterfactual and Discussion

With my fitted model at hand, I am in a position to run different scenarios. In particular, what might be the ideal choice of p for the punishment function by a policymaker?

Ideally, a policymaker would be able to prevent fraud outright, which in my model would mean having direct influence over physicians' choice of $\bar{\kappa}_j$, namely placing a lower bound $\underline{\kappa}$ on it, below which no patient may receive sick leave. In a way, this is already the status quo, except that it cannot be enforced. The "overdetermination" of roles that Markussen and Røed (2017) describe not only leaves physicians serving opposing financial interests (their own and the taxpayers'), but also lets them cover their tracks by misreporting their patients' conditions, which makes monitoring them much harder. As I mentioned, in my setting higher scrutiny *does* apply to sick leaves of 30 days or more, which are automatically sent to the overseer for approval. This is not the case for the routine conditions that make up most of my sample, nor would it be practical.

This isn't necessarily a question of deliberate fraud either. Carlsen et al. (2020) show that under all but the most extreme parameter values, physicians have no incentive to distrust their patients' self-reports, whether or not they suspect feigned symptoms. My preferred policy would therefore seek to align the physician's incentives with the policymaker's, instead of targeting only intentional fraud, so that physicians police themselves by scrutinizing diagnoses more closely.

Suppose for a moment that the policymaker *were* able to place restrictions on $\bar{\kappa}_j$. Under my estimated parameters, if every physician's threshold were raised to at least a fourth of the upper bound, $\underline{\kappa} = 1/4\,\bar{\kappa}_{\max}$, total issuance would fall by 32%. The patients cut off would be those with the lowest medical necessity κ_i among those previously receiving sick leave.

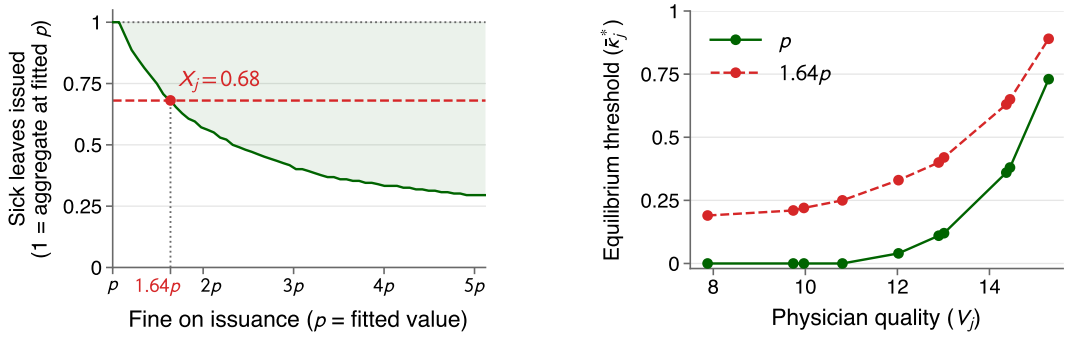
How might the policymaker achieve this through feasible means? The punishment function $P(\cdot)$ is already in the overseer's hands. Since misdiagnosis isn't observed, punishment depends on the sick leaves a physician issues over a given period (X_j), through both the probability and the size of a fine, which I approximate as $P(X_j) = \frac{p}{2}X_j^2$. The policymaker therefore controls the parameter p.

Figure 6 shows what this lever can achieve. Panel 6a plots the *aggregate* sick leaves issued in the market as p rises from its estimated value. The dashed horizontal line marks the level that direct control over physicians' thresholds would reach, with a minimum threshold of $\underline{\kappa} = \bar{\kappa}_{\max}/4$. Raising p by 64% from its fitted value achieves the same aggregate reduction in issuance.

The two policies coincide not only in the aggregate but also in which patients lose their sick leave. In my model, patients with low medical need sort toward the most lenient physicians, since quality matters little to them. A higher fine forces these physicians to become stricter, and Panel 6b shows that, in general equilibrium, every physician raises her threshold in turn. Since no representative physician sets a threshold below that of the most lenient one, bin 1, its threshold acts as an effective lower bound on κ. Raising p thus has an equivalent effect on issuance to imposing some level of $\underline{\kappa}$ directly.

This clean result is driven by some stylized modeling choices. It assumes that patients

Figure 6: Aggregate sick leaves and the punishment parameter



(a) Aggregate sick leaves $\sum_j X_j$

(b) Equilibrium thresholds $\bar{\kappa}_j^*$

Notes: Panel (a): aggregate sick leaves $\sum_j X_j$ in the fitted model as the punishment parameter rises from its estimated value p, normalized to 1 at p. The dashed red line is the level $X' = 0.68X$ reached when every threshold is set to $\underline{\kappa} = \bar{\kappa}_{\max}/4$, which the model attains at $1.64p$. Panel (b): equilibrium thresholds by physician quality at p and at $1.64p$.

place positive probability only on physicians they would be willing to visit, a property I have called free disposal, and that search is costless. As a result, there are no deadweight losses from physician-patient matches that don't result in a visit. These assumptions serve to highlight the model's main mechanisms, but relaxing them would make for more realistic, policy-relevant counterfactuals.

I have also not addressed welfare. Inducing stricter physicians reduces patient visits, which creates a trade-off: patients and physicians lose utility, while taxpayers save on the cost of covered sick leave. Striking the right balance between these forces would require extending the calibration to put a cardinal value on agents' utility and on the fiscal cost of sick leave. I leave this to future work.

7 Conclusion

I have developed and estimated a model of the market for medical treatment and sick leave, in which patients search imperfectly across physicians who differ in quality and choose how leniently to issue sick leave. Competition for patients leads lower-quality physicians to relax their issuance thresholds, producing assortative matching between lenient physicians and patients with low medical need. These patterns arise endogenously from the choices of physicians and patients, under a parsimonious set of modeling assumptions. The model's main insight is that, because of assortative matching, fining physicians on their aggregate issuance is an effective way to deny sick leave to the patients with the least medical need, even though individual fraud is unobservable (in my calibration, raising the fine by 64% cuts sick leaves by 32%).

How well the counterfactual predicts physicians' response to a higher fine depends on two

things: the functional forms I chose for the model's distributions and functions, and the data used to calibrate their parameters. The functional forms are simple and follow precedents in the literature, in particular Schnell (2026). The data are more limited: my moments are the sick leaves issued, overall and of 30 days or more, across ten physician bins, and not every parameter is separately identified, as is the case for μ and σ.

Additional data would strengthen the estimation, either by adding variation across physicians or by allowing more parameters to be set outside the SMM exercise. Two sources stand out. The first is information on each physician's patient base, which would provide an empirical counterpart to Q_j. The second is an independent measure of physician quality, for instance by relating observable characteristics such as education, tenure, and place of work to disaggregated data on salaries and visit revenue. Such data would also allow r_j and τ_j to vary across physicians rather than being common to all of them.

Despite these limitations, the proposed models show how leniency and assortative matching can arise endogenously from competition between physicians and patients' search, and provide a setting with rich predictions that are relevant for policymaking.

Appendix

A.1 Strategic Complementarity of $\bar{\kappa}_j$'s (Proof of Proposition 2.3)

I defined the components s_{ij} of S_i as:

$$s_{ij} \equiv g_i(u_{ij}, u_{i,-j})$$

where $g_i(\cdot)$ is a continuous function weakly increasing in the first argument u_{ij}, weakly decreasing in $u_{i,-j}$. My defining s_{ij} this way has two interlinked corollaries:

Corollary A.1.

$$s_{ij} \mid \kappa_i < \bar{\kappa}_j \quad \leq \quad s_{ij} \mid \kappa_i \geq \bar{\kappa}_j \quad \textit{for } \gamma_i \geq 0 \textit{ (with strict inequality if } \gamma_i > 0\textit{).}$$

Proof. For a fixed value of V_j and κ_i, for all i, the value of $U_i(V_j, \bar{\kappa}_j)$ is $V_j\kappa_i - \tau_j$ if $\kappa_i < \bar{\kappa}_j$, and $\gamma_i + V_j\kappa_i - \tau_j$ if $\kappa_i \geq \bar{\kappa}_j$. For $\gamma_i \geq 0$, u_{ij} is therefore weakly higher if $\kappa_i \geq \bar{\kappa}_j$, and since s_{ij} is weakly increasing in u_{ij}, the corollary follows. □

Corollary A.2.

$$\frac{\Delta s_{ij}}{\Delta \bar{\kappa}_j} \leq 0 \qquad\qquad \frac{\Delta s_{ij}}{\Delta \bar{\kappa}_l} \geq 0, \forall l \neq j$$

Proof. From the argumentation in corollary 1 follows that u_{ij} is weakly decreasing in $\bar{\kappa}_j, \forall j$. Take some $l \neq j$, then s_{ij} is defined in turn as weakly decreasing in u_{il}, which implies it is increasing in $\bar{\kappa}_l$. □

Both corollaries hinge upon my definition of $U_i(\cdot)$ as a step function over κ_i, such that it is discontinuous at $\kappa_i = \bar{\kappa}_j$, where there's a discrete jump of magnitude $\gamma_i \geq 0$. I use them in the next proposition.

Proposition A.3.

$$\frac{\partial Q_j}{\partial \bar{\kappa}_j}, \frac{\partial X_j}{\partial \bar{\kappa}_j} \leq 0 \qquad\qquad \frac{\partial Q_j}{\partial \bar{\kappa}_l}, \frac{\partial X_j}{\partial \bar{\kappa}_l} \geq 0, \forall l \neq j$$

Proof. Recall that patient demand for physician j is:

$$Q_j(\bar{\kappa}_j, \bar{\kappa}_{-j}) \;=\; \int_0^\infty \int_0^\infty s_{ij}(\kappa, \gamma)\, dG(\gamma)\, dF(\kappa)$$

Were physician j to increase her threshold from $\bar{\kappa}_j$ to $\bar{\kappa}_j + \epsilon$, with $\epsilon > 0$, for the mass of patients whose $\kappa_i \in [\bar{\kappa}_j, \bar{\kappa}_j + \epsilon)$, utility would weakly decrease and thus so would their s_{ij}, falling to $s_{ij}^- \leq s_{ij}$ (see first corollary, think of s_{ij}^- as $s_{ij} \mid \kappa_i < \bar{\kappa}_j$ as opposed to $s_{ij} \mid \kappa_i \geq \bar{\kappa}_j$).

The difference that would make for patient demand *ceteris paribus* given the $\bar{\kappa}_{-j}$ strategies of the other physicians (omitted in my notation) is:[14]

$$\Delta Q_j = Q_j(\bar{\kappa}_j + \epsilon) - Q_j(\bar{\kappa}_j) = \int_{\bar{\kappa}_j}^{\bar{\kappa}_j + \epsilon} \int_0^\infty \{s_{ij}^- - s_{ij}\}\, dG(\gamma)\, dF(\kappa)$$

By the definition of a partial derivative:[15]

$$\begin{aligned} \frac{\partial Q_j}{\partial \bar{\kappa}_j} &= \lim_{\epsilon \to 0} \frac{\int_{\bar{\kappa}_j}^{\bar{\kappa}_j + \epsilon} \int_0^\infty \{s_{ij}^- - s_{ij}\}\, dG(\gamma)\, dF(\kappa)}{\epsilon} \\ &= \int_0^\infty \{s_{ij}^-(\bar{\kappa}_j, \gamma) - s_{ij}(\bar{\kappa}_j, \gamma)\}\, dG(\gamma)\, f(\bar{\kappa}_j) \end{aligned}$$

where the derivative is negative if s_{ij}^- is strictly lower than s_{ij}, meaning a shift in leniency would affect the utility and visit probabilities of some positive mass of patients around $\bar{\kappa}_j$; or null, if $s_{ij}^- = s_{ij}$.

In a similar vein, consider some physician $l \neq j$ shifting her threshold from $\bar{\kappa}_l$ to $\bar{\kappa}_l + \epsilon$, with $\epsilon > 0$. The fall in patient visit probabilities in $[\bar{\kappa}_l, \bar{\kappa}_l + \epsilon)$ from s_{il} to $s_{il}^- \leq s_{il}$ could provide a windfall for j (and all other physicians), raising s_{ij} to $s_{ij}^+ \geq s_{ij}$ (see previous corollary). In that case:

$$\Delta Q_j = \int_{\bar{\kappa}_l}^{\bar{\kappa}_l + \epsilon} \int_0^\infty \{s_{ij}^+ - s_{ij}\}\, dG(\gamma)\, dF(\kappa)$$

[14] Besides simplifying $Q_j(\bar{\kappa}_j, \bar{\kappa}_{-j})$ into $Q_j(\bar{\kappa}_j)$, I also forgo the arguments of $s_{ij}(k, \gamma)$, as I've done many times already, to avoid notation clutter.

[15] The notation for $s_{ij}^-(\bar{\kappa}_j, \gamma)$ is to be taken to mean the visit probability $s_{ij}^- \leq s_{ij}$ of the mass of patients at different levels of γ atomized at $\kappa_i = \bar{\kappa}_j$.

thus rendering:

$$\begin{aligned}\frac{\partial Q_j}{\partial \bar{\kappa}_l} &= \lim_{\epsilon \to 0} \frac{\int_{\bar{\kappa}_l}^{\bar{\kappa}_l + \epsilon} \int_0^\infty \{s_{ij}^+ - s_{ij}\}\, dG(\gamma)\, dF(\kappa)}{\epsilon} \\ &= \int_0^\infty \{s_{ij}^+(\bar{\kappa}_l, \gamma) - s_{ij}(\bar{\kappa}_l, \gamma)\}\, dG(\gamma)\, f(\bar{\kappa}_l)\end{aligned}$$

As for X_j, the steps are almost the same, except that after raising the threshold by ϵ the sick leaves granted around $\bar{\kappa}_j$ go to 0, giving:

$$\frac{\partial X_j}{\partial \bar{\kappa}_j} = -\int_0^\infty s_{ij}(\bar{\kappa}_j, \gamma)\, dG(\gamma)\, f(\bar{\kappa}_j)$$

such that it is simply required that j has positive demand for the mass of patients whose $\kappa_i = \bar{\kappa}_j$ for this derivative to be strictly negative, else it is null.

Likewise, a rise in $\bar{\kappa}_l$ raises s_{ij} for patients at $\bar{\kappa}_l$, who are granted sick leave by j only if $\bar{\kappa}_l \geq \bar{\kappa}_j$:

$$\frac{\partial X_j}{\partial \bar{\kappa}_l} = \mathbb{1}[\bar{\kappa}_l \geq \bar{\kappa}_j] \int_0^\infty \{s_{ij}^+(\bar{\kappa}_l, \gamma) - s_{ij}(\bar{\kappa}_l, \gamma)\}\, dG(\gamma)\, f(\bar{\kappa}_l) \;\geq\; 0$$

□

Before dealing with Proposition 2.3 I need one last intermediate result.

Proposition A.4.

$$\frac{\partial^2 Q_j}{\partial \bar{\kappa}_j \partial \bar{\kappa}_l} = 0 \quad \textit{and} \quad \frac{\partial^2 X_j}{\partial \bar{\kappa}_j \partial \bar{\kappa}_l} = 0\,, \quad \forall j \textit{ and } l,\, \bar{\kappa}_j \neq \bar{\kappa}_l$$

Proof. My proof will be heuristic. Consider

$$\frac{\partial Q_j}{\partial \bar{\kappa}_j} = \int_0^\infty \{s_{ij}^-(\bar{\kappa}_j, \gamma) - s_{ij}(\bar{\kappa}_j, \gamma)\}\, dG(\gamma)\, f(\bar{\kappa}_j)$$

Patient visit probabilities s_{ij} are taken at the atom $\kappa_i = \bar{\kappa}_j$. For any other physician l such that $\bar{\kappa}_l \neq \bar{\kappa}_j$, an infinitesimally small change to $\bar{\kappa}_l$ doesn't affect the utility of patients at $\bar{\kappa}_j$, who will remain above or below l's threshold as before. Thus their visit probability s_{ij} also remains unaffected. If $\bar{\kappa}_l$ *does* equal $\bar{\kappa}_j$, then the following limit is not properly defined:

$$\lim_{\epsilon \to 0} \frac{\frac{\partial Q_j(\kappa_l + \epsilon)}{\partial \bar{\kappa}_j} - \frac{\partial Q_j(\kappa_l)}{\partial \bar{\kappa}_j}}{\epsilon}$$

because as ϵ approaches 0 from the right the limit is weakly positive (≥ 0), whereas approaching from the left it's strictly 0. Hence, the derivative $\frac{\partial^2 Q_j}{\partial \bar{\kappa}_j \partial \bar{\kappa}_l}$ does not exist for j and $l \neq j$ when $\bar{\kappa}_j = \bar{\kappa}_l$.

□

Proof of Proposition 2.3.

Consider two physicians j, l, such that $\bar{\kappa}_j \neq \bar{\kappa}_l$. If I break down the mixed partial derivative of U_j over $\bar{\kappa}_j$ and $\bar{\kappa}_l$, I get:

$$\begin{aligned}\frac{\partial^2 U_j}{\partial\bar{\kappa}_j\partial\bar{\kappa}_l} =& \underbrace{R_j''(Q_j(\bar{\kappa}_j,\bar{\kappa}_{-j}))}_{\leq 0}\cdot\underbrace{\frac{\partial Q_j}{\partial\bar{\kappa}_l}}_{\geq 0}\cdot\underbrace{\frac{\partial Q_j}{\partial\bar{\kappa}_j}}_{\leq 0}+\underbrace{R_j'(Q_j(\bar{\kappa}_j,\bar{\kappa}_{-j}))}_{\geq 0}\underbrace{\frac{\partial^2 Q_j}{\partial\bar{\kappa}_j\partial\bar{\kappa}_l}}_{=0}\\ &-\underbrace{P''(X_j(\bar{\kappa}_j,\bar{\kappa}_{-j}))}_{\geq 0}\cdot\underbrace{\frac{\partial X_j}{\partial\bar{\kappa}_l}}_{\geq 0}\cdot\underbrace{\frac{\partial X_j}{\partial\bar{\kappa}_j}}_{\leq 0}-\underbrace{P'(X_j(\bar{\kappa}_j,\bar{\kappa}_{-j}))}_{\geq 0}\underbrace{\frac{\partial^2 X_j}{\partial\bar{\kappa}_j\partial\bar{\kappa}_l}}_{=0}\\ =& \underbrace{R_j''(Q_j(\bar{\kappa}_j,\bar{\kappa}_{-j}))\cdot\frac{\partial Q_j}{\partial\bar{\kappa}_l}\cdot\frac{\partial Q_j}{\partial\bar{\kappa}_j}}_{\geq 0}\underbrace{-P''(X_j(\bar{\kappa}_j,\bar{\kappa}_{-j}))\cdot\frac{\partial X_j}{\partial\bar{\kappa}_l}\cdot\frac{\partial X_j}{\partial\bar{\kappa}_j}}_{\leq 0} \qquad \geq 0\end{aligned}$$

and a non-negative mixed derivative of physicians' utility over their strategy is the definition of strategic complementarity.

□

A.2 Optimal Reservation Utility (Proof of Equation 7)

I follow the argument of Ljungqvist and Sargent (2012). Let for the moment $\tilde{F}_i(U) = \Pr\{u_{ij} \leq U\}$ be the distribution of utility patient i receives from doctor visits, with $\tilde{F}_i(0) = 0$, $\tilde{F}_i(B) = 1$, for $B < \infty$. Each period, the patient draws an 'offer' U from $\tilde{F}_i$, which is to say, his search led him to a physician j such that $u_{ij} = U$. He may either accept and have physician j assigned to him, receiving utility U per period henceforth, or look for a new physician next period.

Let u_t be the patient's utility for period t, where it's 0 if he doesn't visit a physician and U if he has accepted an assigned physician which renders him utility $u_{ij} = U$. This patient will seek to maximize $\mathbb{E}[\sum_{t=0}^{\infty}\beta^t u_t]$, where $0 < \beta < 1$ is his discount factor.

Let $V(U)$ be the expected value of $\sum_{t=0}^{\infty}\beta^t u_t$ of a patient with an offer U in hand. Assuming no recall, under optimal behavior the value function $V(U)$ satisfies the Bellman equation:

$$V(U) = \max\left\{\frac{U}{1-\beta},\ \ \beta\int V(U')\,d\tilde{F}_i(U')\right\} \tag{12}$$

The patient chooses an optimal value $\bar{U}_i$, such that offers $U \geq \bar{U}_i$ are accepted (the visit takes place). As such, the solution takes on the following form:

$$V(U) = \begin{cases}\frac{\bar{U}_i}{1-\beta} + \beta\int V(U')\,d\tilde{F}_i(U') & \text{if}\quad U \leq \bar{U}_i,\\ \frac{U}{1-\beta} & \text{if}\quad U \geq \bar{U}_i.\end{cases} \tag{13}$$

Using equation (13), I can convert the functional equation (12) into an ordinary equation in the reservation utility $\bar{U}_i$. Evaluating $V(\bar{U}_i)$ and using equation (13), I have:

$$\frac{\bar{U}_i}{1-\beta} = \beta \int_0^{\bar{U}_i} \frac{\bar{U}_i}{1-\beta}\, d\tilde{F}_i(U') + \beta \int_{\bar{U}_i}^{B} \frac{U'}{1-\beta}\, d\tilde{F}_i(U')$$

or

$$\begin{aligned} &\frac{\bar{U}_i}{1-\beta} \int_0^{\bar{U}_i} d\tilde{F}_i(U') + \frac{\bar{U}_i}{1-\beta} \int_{\bar{U}_i}^{B} d\tilde{F}_i(U') \\ &\quad = \beta \int_0^{\bar{U}_i} \frac{\bar{U}_i}{1-\beta}\, d\tilde{F}_i(U') + \beta \int_{\bar{U}_i}^{B} \frac{U'}{1-\beta}\, d\tilde{F}_i(U') \end{aligned}$$

or

$$\bar{U}_i \int_0^{\bar{U}_i} d\tilde{F}_i(U') = \frac{1}{1-\beta} \int_{\bar{U}_i}^{B} \left(\beta U' - \bar{U}_i\right) d\tilde{F}_i(U').$$

Adding $\bar{U}_i \int_{\bar{U}_i}^{B} d\tilde{F}_i(U')$ to both sides gives

$$\bar{U}_i = \frac{\beta}{1-\beta} \int_{\bar{U}_i}^{B} \left(U' - \bar{U}_i\right) d\tilde{F}_i(U') \tag{14}$$

Finally, for my purposes, I discretize $\int_{\bar{U}_i}^{B} \left(U' - \bar{U}_i\right) d\tilde{F}_i(U')$ as a summation over the countable set of J doctors. The discrete equivalent of equation (14) is:

$$\bar{U}_i = \frac{\beta}{1-\beta} \frac{\sum_{j=1}^{J} \mathbb{1}[u_{ij} \geq \bar{U}_i] \cdot (u_{ij} - \bar{U}_i)}{\sum_{j=1}^{J} \mathbb{1}[u_{ij} \geq \bar{U}_i]} \tag{7}$$

This is the condition that characterizes patient thresholds in the explicit search model.

A.3 Parameter Sources

Physician quality by bins (V_1 ... V_{10})

As mentioned in the main text, three criteria are considered for the *quality score*:

i Average duration of sick leaves issued.

ii Percentage of sick leaves of 30+ days over total in sampled period.

iii Amount of sick leaves of 30+ days issued.

For each of the criteria the *percentile rank* which each physician occupies is recorded, all in increasing order, i.e. from lowest to highest average duration, from lowest to highest percentage/amount of $\geq$ 30 days sick leaves. Call the percentile rank which physician j occupies in each category p_j^i, p_j^{ii} and p_j^{iii}. The *quality score* of physician j is the simple average of these three, that is:

$$\text{score}_j \;=\; \frac{1}{3}p_j^i \;+\; \frac{1}{3}p_j^{ii} \;+\; \frac{1}{3}p_j^{iii}$$

Table 3: Most common conditions and matched fees

cie10 cat.	Instances	%	Informal Description	Matched Field	Full Price	Aff. Fee
M54	379,332	0.20	Back pain	Traumatology	$ 18,130	$ 7,250
F32	304,399	0.16	Depression	Psychiatry	$ 27,510	$ 11,000
F41	284,129	0.15	Anxiety	Psychiatry	$ 27,510	$ 11,000
F43	244,468	0.13	Stress	Psychiatry	$ 27,510	$ 11,000
J20	191,615	0.10	Bronchitis	General	$ 14,270	$ 7,440
A09	186,535	0.10	Gastroenteritis	General	$ 14,270	$ 7,440
empty	152,239					
M75	115,585	0.06	Shoulder	Traumatology	$ 18,130	$ 7,250
J00	92,660	0.05	Common cold	General	$ 14,270	$ 7,440
M51	54,800	0.03	Intervertebral disc	Traumatology	$ 18,130	$ 7,250
K52	49,182	0.03	Gastroenteritis	General	$ 14,270	$ 7,440

Notes: The ten most frequent *cie10* categories reported in the sample, plus sick leaves with no category (*empty*). The % column is each category's share among the ten. Full Price and Aff. Fee are in Chilean pesos, from FONASA's fee schedule for the Free Choice Modality.

The quality bins are then set up from the decile ranks of physicians in their *quality score*. Bin 1, for example, is simply the first decile rank of q_score, the lowest 10% of physicians according to their quality score. Each bin is therefore equally sized. The parameters V_1 ... V_{10} associated with each bin are then estimated in my SMM exercise, constrained to be increasing in quality. Simply put: $V_{10} > V_9 > ... > V_1$.

Revenue by visit (r_j) and cost of visit (τ_j)

I use a single r and τ for all physicians to simplify the analysis. I select the ten most frequent $cie10$ categories, the international standard classification of medical conditions. These ten, as well as non-categorized conditions (labelled *empty* in the Table), account for 47.91% of all sick leaves in my sample. I then match the conditions to a particular field of practice. Strictly speaking, physicians can issue sick leave outside their own field, but this is a useful rule of thumb.

From FONASA's fee schedule (2024), I take, for each field, the full price of a visit and the fee paid by FONASA affiliates under the Free Choice Modality (MLE). These are the empirical counterparts of the physician's revenue per visit r and the patient's cost of visit τ, respectively.

My final value for r and τ is the weighted average of these prices, weighted by the relative frequency of each category and its matched full and benefit price. This amounts to taking the dot product of the '%' column with the 'Full Price' and 'Aff. Fee' columns in Table 3.

The resulting values are $r = 21{,}182$ and $\tau = 8{,}944$. For estimation, I express them in thousands of pesos (e.g., 8.944 instead of 8,944). Table 3 shows the values used.